

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 2.3: Chemical Bonding — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 2.3: Chemical Bonding
Driving Question	DRIVING QUESTION: Why do the aluminium sufuria, the graphite pencil and the table salt on our kitchen table look similar as solids but behave so differently — and what is happening between the atoms inside each one that controls every property we can measure?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Three Kitchen Mysteries: Anchoring the Phenomenon & Eliciting Prior Knowledge	Students observed a live conductivity demonstration using a 1.5 V battery-and-bulb circuit with three Kenyan household items: table salt (from the coast), aluminium sufuria wire, and a graphite rod from a used AA battery. They saw that aluminium conducted immediately, dry salt did not conduct, dissolved salt did conduct, and — most surprisingly — graphite also conducted electricity even though it is a non-metal.	Students surfaced and documented their prior knowledge and misconceptions about electrical conductivity, melting points, and mechanical properties of solids. They learned that their existing rule ('only metals conduct') is insufficient to explain all the evidence, and they identified specific gaps in their understanding that the unit will address.	No explanations of bond types or particle structure were provided in this lesson. The lesson's purpose was to open the inquiry, not close it. The three items remain mysterious. Students have committed their initial predictions and initial models to their science journals — these will be revised and explained across Lessons 2–8 as each bond type (ionic, metallic, covalent/giant covalent) is introduced.
Lesson 2 Predicting the Invisible: What Are Our Three Kitchen Items Made Of?	Three kitchen items — a lump of Kenyan coastal salt (NaCl), a length of aluminium sufuria wire, and a graphite rod from a used AA battery — displayed on the demonstration desk. No testing was done in this lesson; learners observed the items visually and by recall only.	Learners activated prior knowledge of valence electrons and electron arrangement from Sub-Strand 2.1. They practised writing structured predictions using an 'I think ... because ...' frame, connecting macroscopic properties (conductivity, melting point, mechanical strength) to particle-level reasoning about how atoms might share or transfer electrons.	Nothing has been formally explained yet — this lesson is intentionally the PREDICT phase. All predictions remain on the Driving Question Board as hypotheses to be tested. The class driving question was formally introduced: 'Why do these three solids behave so differently, and what is happening between the atoms inside each one?'
Lesson 3 Ionic Bonding: Why Does Table Salt Conduct Only When Dissolved?	Dry NaCl crystals placed in a conductivity circuit did NOT light the bulb. Distilled water did NOT light the bulb. NaCl dissolved in water DID light the bulb. A dot-and-cross	Ionic bonds form by the complete transfer of one or more electrons from a metal atom to a non-metal atom, creating oppositely charged ions (cations and anions). In solid	The coastal table salt on our demonstration desk does not conduct electricity as a solid because its Na^+ and Cl^- ions are locked in fixed positions in a giant ionic lattice — they

	diagram showed the Na atom donating its single valence electron to the Cl atom, forming Na^+ and Cl^- ions with complete octets. A 2-D lattice model built from two-coloured maize grains showed alternating positive and negative ions in a regular pattern; shifting one row caused same-charge ions to become neighbours.	NaCl, Na^+ and Cl^- ions are held in a giant ionic lattice by strong electrostatic forces acting in all directions. These forces require very high temperatures to overcome, giving NaCl its high melting point ($801\text{ }^\circ\text{C}$). In the solid state, ions are locked in fixed positions and cannot carry charge, so solid NaCl does not conduct electricity. When NaCl dissolves in water or is melted, the ions become free to move, allowing electrical conduction. The lattice structure also explains brittleness: a mechanical blow shifts an ion layer so that like-charged ions align, creating repulsive forces that split the crystal.	cannot move to carry charge. When the salt dissolves in water (as in a cup of githeri broth), the lattice breaks apart and the ions become free to migrate to the electrodes, completing the circuit and lighting the bulb. The same lattice — with its strong electrostatic forces acting throughout the entire three-dimensional structure — also explains why salt requires over $800\text{ }^\circ\text{C}$ to melt, far beyond what a home jiko can achieve. The salt's brittleness (it shatters rather than bends) is explained by the layer-shift repulsion in the lattice model. This fully accounts for the salt observations in our anchoring phenomenon. The aluminium and graphite items on the desk still need a different type of bonding to explain their behaviour — that is the question now written in red on our Driving Question Board.
Lesson 4 Valence Electrons, Dot-and-Cross Diagrams, and the Six Bond Types	Students observed (via teacher board demonstration) how valence electrons behave during bond formation: Na transfers its single valence electron to Cl, producing oppositely charged ions held by electrostatic attraction; H, O, and other non-metals share electron pairs to form covalent bonds. They also observed dot-and-cross diagrams for H_2, H_2O, and O_2 and practised drawing diagrams for assigned molecules (NH_3, Cl_2, HCl, MgO, CO_2) in pairs.	Students learned the definitions and distinguishing features of all six bond types — ionic (electron transfer, e.g., coastal NaCl), covalent (electron sharing, e.g., O_2 and H_2O), dative-covalent (both shared electrons from one atom, e.g., NH_4^+ in DAP fertiliser), hydrogen bonds (H bonded to N/O/F, e.g., water in Lake Victoria), metallic (delocalised electron sea, e.g., aluminium sufuria), and Van der Waals forces (temporary induced dipoles, e.g., graphite layers and iodine crystals). They learned to connect each bond type to expected properties (conductivity, melting point, mechanical behaviour).	Students explained, through revised particle-level sketches and written sentences, how the bond type inside each kitchen item accounts for one observable property from the anchoring phenomenon: the electrostatic lattice of Na^+Cl^- explains the salt's high melting point and conductivity only in solution; the sea of delocalised electrons in aluminium explains solid-state conductivity and malleability; and the combination of strong covalent bonds within graphite layers and weak Van der Waals forces between layers explains both graphite's electrical conductivity (delocalised electrons within layers) and its ability to flake off onto paper.
Lesson 5 Practical Investigation of Physical Properties — Connecting Evidence to the Anchoring Phenomenon	Learners observed that: NaCl crystals do not conduct electricity as a solid but dissolve in water; candle wax does not dissolve in water and does not conduct; graphite conducts electricity as a solid despite being a non-metal; aluminium conducts electricity as a solid and is flexible and malleable rather than brittle. Learners also observed that candle wax is soft and easily scratched, NaCl is hard, graphite is	Learners learned that the physical properties of a solid substance — its solubility, electrical conductivity and mechanical behaviour — are directly determined by the type of bonding and structure holding its particles together. They learned that conductivity in a solid requires particles (electrons or ions) that are free to move, and that this freedom depends entirely on bond type: metallic and layered	Learners can now explain the anchoring phenomenon more completely than in Lesson 1: aluminium conducts and bends because delocalised electrons move freely through the metallic lattice and the layers of metal ions can slide past each other; graphite conducts because delocalised electrons within each carbon layer are free to move along it and carry charge, even though carbon is a non-metal; dry NaCl

	brittle and snaps, and aluminium bends without breaking.	giant atomic structures allow electron movement, while giant ionic and simple molecular structures do not (in the solid state).	does not conduct because its ions are locked in a rigid lattice and cannot carry charge until the lattice is broken by dissolving or melting; candle wax neither conducts nor dissolves because its molecules are held together only by weak Van der Waals forces and contain no ions or free electrons. The driving question has advanced: learners now have experimental evidence, not just theoretical predictions, to support bond-type explanations of physical properties.
Lesson 6 Seeing the Invisible: Drawing Ionic and Covalent Bonds with Dot-and-Cross Diagrams	Using a virtual laboratory simulation (projected or on individual devices), learners observed animated ionic bond formation in NaCl and MgO — watching sodium and magnesium transfer electrons to chlorine and oxygen respectively — and covalent bond formation in HCl, NH ₃ and CH ₄ — watching atoms share electron pairs without transferring charge.	Ionic bonding involves complete electron transfer from a metal to a non-metal, producing oppositely charged ions (e.g., Na ⁺ and Cl ⁻ in table salt from the Kenyan coast); covalent bonding involves electron sharing between non-metals, producing neutral molecules (e.g., CH ₄ in cooking gas, NH ₃). Learners can now draw accurate dot-and-cross diagrams for all five compounds and explain why ionic compounds produce mobile ions when dissolved (connecting to the salt conductivity observation in the anchoring phenomenon) while simple covalent molecules do not.	The dot-and-cross diagrams explain why dry table salt does not conduct electricity (ions are locked in a rigid lattice — no mobile charge carriers) but conducts when dissolved in water (ions become free to move). They also explain why CH ₄ and similar simple covalent molecules have low melting points (only weak intermolecular forces — explored further in Lesson 7). The aluminium sufuria and graphite rod remain in the 'still unexplained' column of the DQB, motivating Lesson 7's investigation of metallic and giant covalent structures.
Lesson 7 Bond Types, Structures and Physical Properties: Making Sense of the Data	We tested and recorded data on the conductivity, melting points and mechanical behaviour of table salt (NaCl from the Kenyan coast), aluminium (from a sufuria handle) and graphite (from a used AA battery rod) across Lessons 1–6. We observed that dry salt does not conduct but salt solution does; aluminium conducts as a solid and bends without breaking; graphite conducts as a solid but snaps rather than bends, and leaves marks on paper.	Each substance has a different structural type determined by its bond type: NaCl forms a giant ionic lattice where ions are fixed in position (so no conductivity in solid state, but high melting point); aluminium forms a giant metallic lattice with a sea of delocalised electrons (so conducts in solid state, is malleable, and has a moderate melting point); graphite forms a giant covalent layered structure where each carbon has one delocalised electron free to move along the layer (so conducts), and weak Van der Waals forces between layers mean layers slide off easily (so it is brittle across layers and marks paper). We can predict the properties of any substance if we know its structural type.	The three kitchen items look similar as solids but behave so differently because their internal particle-level structures — determined by their bond types — are completely different. In NaCl, charged ions locked in a rigid lattice cannot move to carry current (hence no conductivity in solid form) but the strong electrostatic forces holding the lattice together require enormous energy to overcome (hence a very high melting point of 801 °C). In aluminium, metallic bonding produces a 'sea' of freely moving delocalised electrons that carry charge immediately (hence solid conductivity) and a lattice of positive ions whose layers can slide (hence malleability). In graphite, covalent bonding within layers with one free electron per carbon enables conductivity along the layers, while weak Van der Waals forces between layers allow

			them to shear apart onto paper. The bond type controls the structure; the structure controls every property we can measure.
Lesson 8 Final Explanation — Four Structural Types, Jigsaw Presentations and the Complete Story of the Kitchen Items	In Lesson 1 we observed three items from a Kenyan kitchen: a table-salt crystal (NaCl from the coast), an aluminium sufuria handle wire and a graphite rod from a dry-cell battery. We saw that dry NaCl did not conduct electricity but its solution did and it had a very high melting point; aluminium conducted electricity immediately as a solid, was shiny and bent without breaking; and graphite — a non-metal — also conducted electricity yet was brittle and left layers on paper.	Across Lessons 2–8 we learned that the differences between these three substances are explained by four giant structural types and their bonding forces: (1) Giant Ionic (NaCl) — Na^+ and Cl^- ions held in a 3D lattice by strong electrostatic attraction; ions are fixed in the solid so it cannot conduct, but dissolve to give mobile ions that do conduct; high lattice energy means a very high melting point. (2) Giant Metallic (Aluminium) — Al^{3+} cations in a regular lattice surrounded by a 'sea' of delocalised electrons; the electrons carry electrical charge so aluminium conducts as a solid; the cation layers can slide over the electron sea so the metal is malleable and ductile. (3) Giant Atomic / Layered Covalent (Graphite) — carbon atoms covalently bonded in flat hexagonal layers; each carbon uses only three of its four valence electrons for bonding, leaving one delocalised electron per atom that can carry charge along the layer — explaining why graphite conducts; the layers are held together only by weak Van der Waals forces, so they slide apart easily — explaining why graphite is brittle and leaves a pencil mark. We also studied simple molecular structures (e.g. iodine), where discrete molecules are held by weak Van der Waals forces, giving low melting points and poor conductivity.	We can now give a complete, particle-level explanation of the anchoring phenomenon: the three kitchen items look similar as solids but behave so differently because they belong to different structural categories. NaCl is a giant ionic lattice — its high melting point and conductivity only in solution both follow directly from the strength and arrangement of electrostatic bonds between fixed ions. Aluminium is a giant metallic structure — its conductivity as a solid, its shine and its malleability all follow from the presence of freely moving delocalised electrons and the ability of cation layers to slide. Graphite is a giant atomic layered covalent structure — its unexpected conductivity follows from delocalised electrons within each hexagonal layer, while its brittleness and ability to leave marks on paper follow from the weak Van der Waals forces between layers. Bond type controls structure; structure controls every property we can measure.

ATTRIBUTION AND LICENSING

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